Group Topic: **Big Data Analysis for Rainfall-Based Flood Risk Prediction in Albay using SYNOP Data** Week 3

Git Repo/ Colab Link: https://github.com/PotatoKevz/Big-Data-Analysis-2026 Repository Date: May 2,

---

Objectives

The main goal for this week was to identify the most suitable machine learning approach for predicting flood risk in Albay using SYNOP rainfall data. Specifically, the group aimed to:

- Choose the best algorithm among Prediction (Regression), Clustering, and Classification.
- Implement the selected algorithm in Python using actual meteorological data from Station 98444 (Legazpi, Albay) together with data from six nearby stations.
- Improve the model through fine-tuning to make the predictions more accurate and reliable for flood risk assessment.

Tasks Completed

1. Choosing an Algorithm

After evaluating the three approaches, the group decided to use Classification for the project. Three classification models were selected for testing and comparison.

The classifiers used were:

- **Random Forest Classifier** – selected as the main model because it performs well on tabular datasets, handles missing values effectively through imputation, and provides feature importance rankings.
- **Gradient Boosting Classifier** – used as a secondary model for comparison.
- **Logistic Regression** – included as the baseline linear model.

2. Implementation of the Algorithm in Code

The implementation was organized into a structured Python pipeline named flood_risk_predictor.py with the following stages:

- **Data Loading** – All CSV files from the clean_dataset/ folder were loaded and parsed. Datetime columns were standardized, numeric values were converted properly, and each dataset was assigned to its corresponding station ID.
- **Feature Engineering** – For Station 98444 (Albay/Legazpi), the script generated rolling rainfall windows for 12-hour, 24-hour, and 48-hour intervals. Lagged rainfall features using 3-hour, 6-hour, and 12-hour lookback periods were also added together with pressure tendency

columns.

The six supporting stations also contributed their own rainfall and pressure features so the model could learn upstream weather patterns from each location.

- **Label Generation** – Flood risk labels were assigned using a four-class system based on 24-hour accumulated rainfall thresholds aligned with DOST-PAGASA standards: LOW (<15 mm), MODERATE (15–50 mm), HIGH (50–100 mm), and EXTREME (>100 mm).
- **Model Training** – The dataset was split into 80% training and 20% testing using stratification across all risk classes. Pipelines were built using scikit-learn with median imputation and standard scaling when necessary. The models were evaluated using Accuracy, Weighted F1 Score, Precision, Recall, and AUC-OvR. A 5-fold cross-validation was also applied to the Random Forest model.
- **Outlier / Extreme Event Detection** – A separate module was created to detect extreme rainfall events using IQR fencing (Q3 + 1.5 × IQR) and Z-score thresholding ($|z| > 3.0$). Detected events were further grouped into "extreme" and "catastrophic" categories.

3. Fine-Tuning the Algorithm

Several adjustments were made to improve the model's performance and solve issues encountered during the initial testing stage.

- **Class Imbalance** – Since LOW-risk observations heavily dominated the dataset, the Random Forest and Logistic Regression models were configured with class_weight='balanced'. This prevented the models from predicting LOW risk for nearly all observations.
- **Hyperparameter Adjustments** – The Random Forest model was configured with 200 trees, a maximum depth of 12, and min_samples_leaf=5 to reduce overfitting. The Gradient Boosting model used a learning rate of 0.1, 100 estimators, and a maximum depth of 5.
- **Feature Expansion** – Instead of relying only on Albay station data, rolling rainfall features from all six supporting stations were added. This helped the model capture directional weather signals such as incoming typhoon activity from nearby areas like Catanduanes.
- **Validation** – A 5-fold cross-validation process was added to the Random Forest model to evaluate how well the model generalizes beyond a single train-test split.

Tasks to be Completed

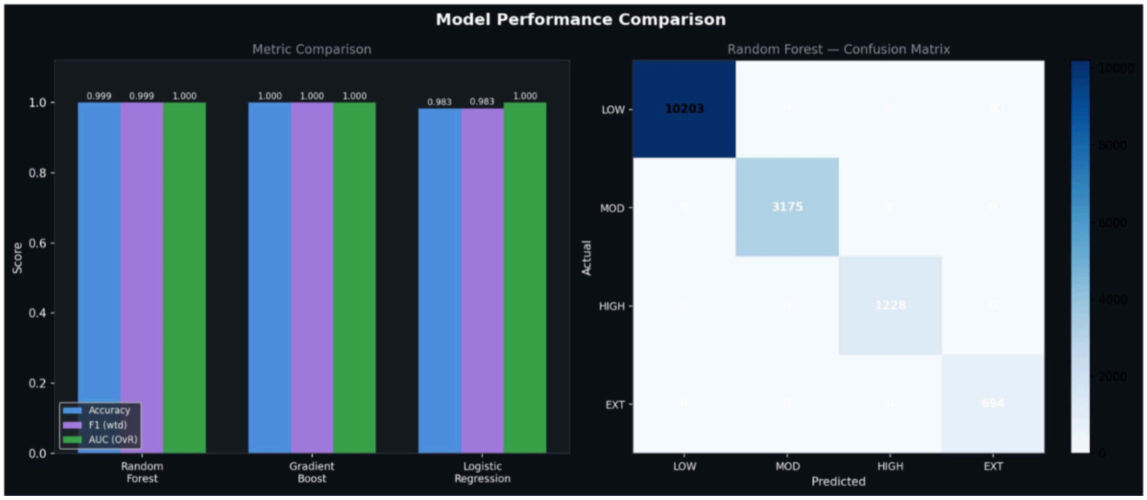
The group plans to complete the additional tasks that will be assigned in the following week.

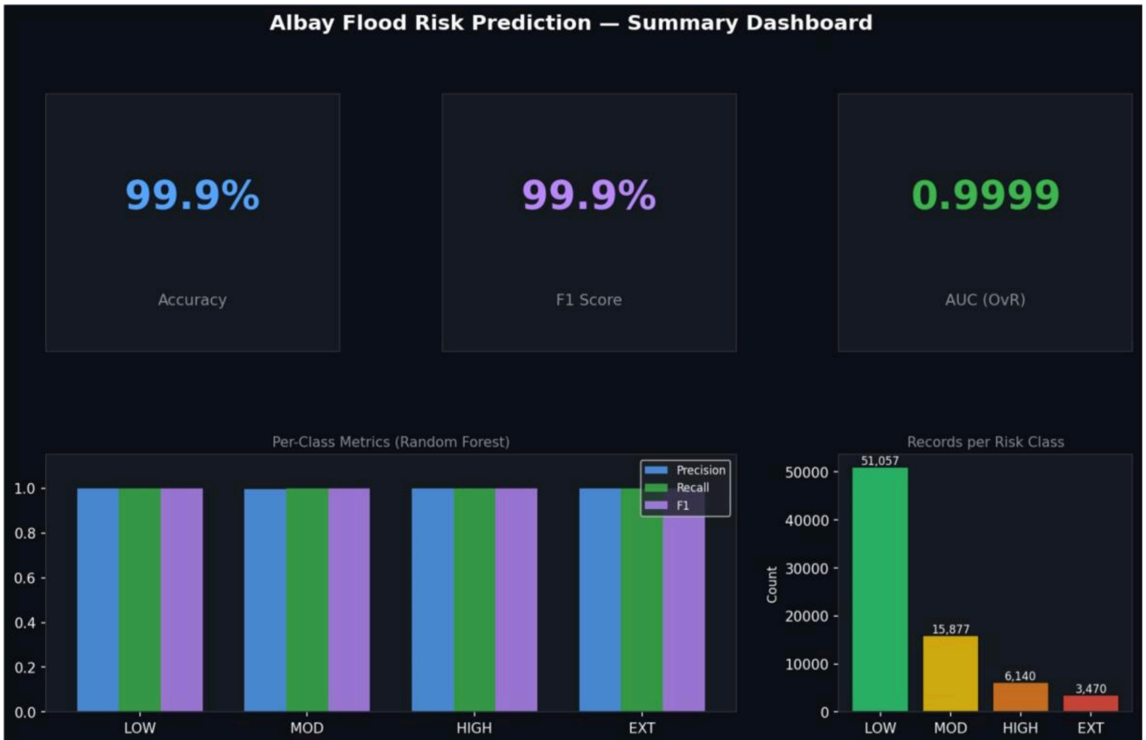
Results and Analysis

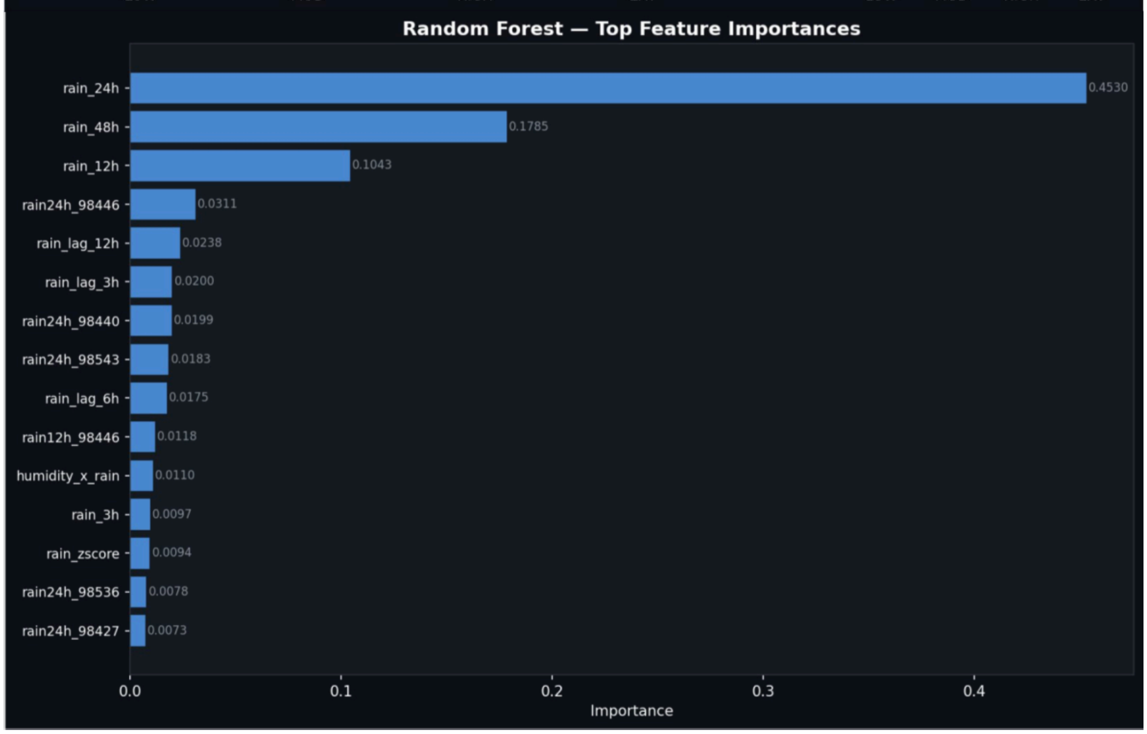
The pipeline was executed using the complete cleaned dataset covering all seven SYNOP stations from 2000 to 2026. The primary station, 98444 (Albay/Legazpi), contributed 76,544 records. The six supporting stations also provided additional context: Camarines Norte (98440) with 76,899 records, Catanduanes (98446) with 76,283, Infanta (98434) with 75,647, Masbate (98543) with 75,956, Romblon (98536) with 73,888, and Tayabas (98427) with 75,034 records.

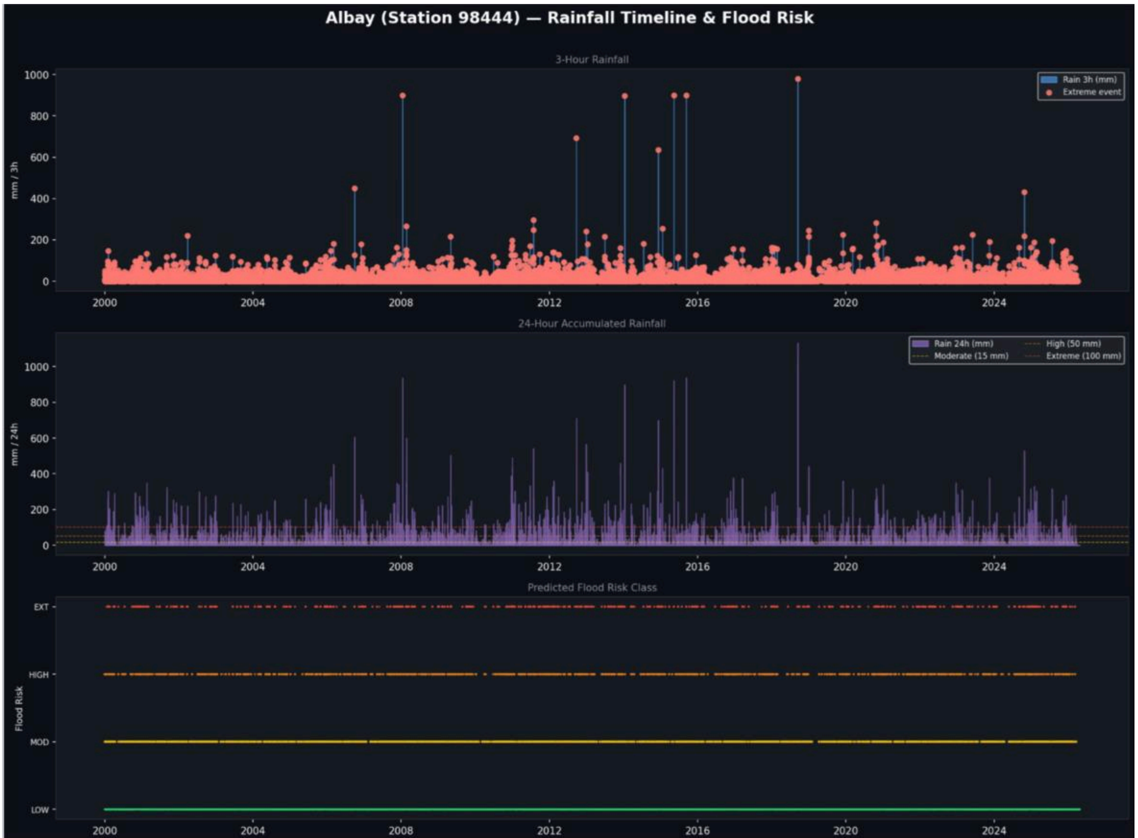
Rainfall, pressure, and humidity readings from the supporting stations were integrated into the model as separate features alongside the Albay station data. For the primary station, the final risk distribution consisted of 51,057 LOW, 15,877 MODERATE, 6,140 HIGH, and 3,470 EXTREME observations. This confirmed the class imbalance issue that required balanced weighting during fine-tuning.

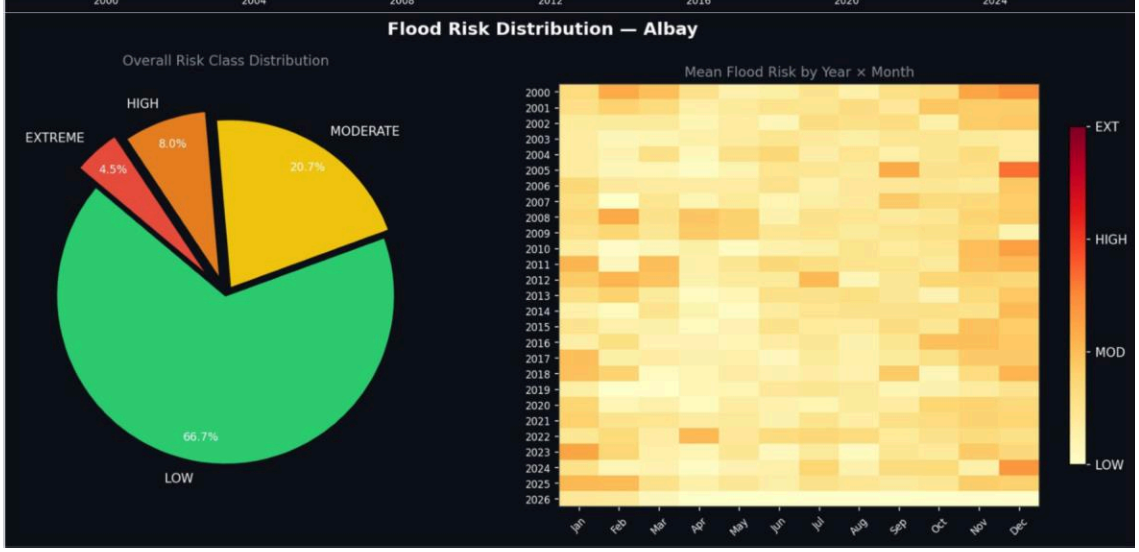
- All three models performed very well. Gradient Boosting achieved the best performance with 99.97% accuracy and a perfect 1.000 AUC score. Random Forest followed closely with 99.94% accuracy and a 0.9999 AUC score. Logistic Regression also performed well with 98.32% accuracy, although it was noticeably lower than the tree-based models. The confusion matrix for Random Forest showed near-perfect classification across all four risk levels, correctly predicting 10,203 LOW, 3,175 MODERATE, 1,228 HIGH, and 694 EXTREME observations with almost no misclassifications.
- Feature importance analysis from the Random Forest model showed that rain_24h was the strongest predictor with an importance score of 0.4530, followed by rain_48h at 0.1785 and rain_12h at 0.1043. The supporting stations also contributed meaningful signals. Catanduanes (98446) had the strongest contribution among them with rain24h_98446 at 0.0311 and rain12h_98446 at 0.0118, showing its value as an upstream typhoon indicator. Camarines Norte (98440) followed with 0.0199, while Masbate (98543) contributed 0.0183. Romblon (98536) and Tayabas (98427) also contributed smaller but still noticeable signals at 0.0078 and 0.0073 respectively. These results confirmed that the multi-station approach was effective because the model learned from regional rainfall patterns rather than depending only on local Albay readings.
- The outlier detection module identified 17,026 extreme rainfall events throughout the dataset. The timeline revealed recurring extreme events across the 2000–2026 period, with the highest spikes reaching almost 1,000 mm within a 3-hour window around 2019–2020. The Z-score visualization also showed several catastrophic outliers far above the 3-sigma threshold, with the highest recorded Z-score exceeding 70.
- The final risk distribution showed that LOW-risk conditions represented 66.7% of all observations, followed by MODERATE at 20.7%, HIGH at 8.0%, and EXTREME at 4.5%. The year-by-month heatmap also showed that flood risk becomes more severe during typhoon season, especially from August to November. Some years such as 2005 and 2008 displayed longer and more intense high-risk periods compared to others.

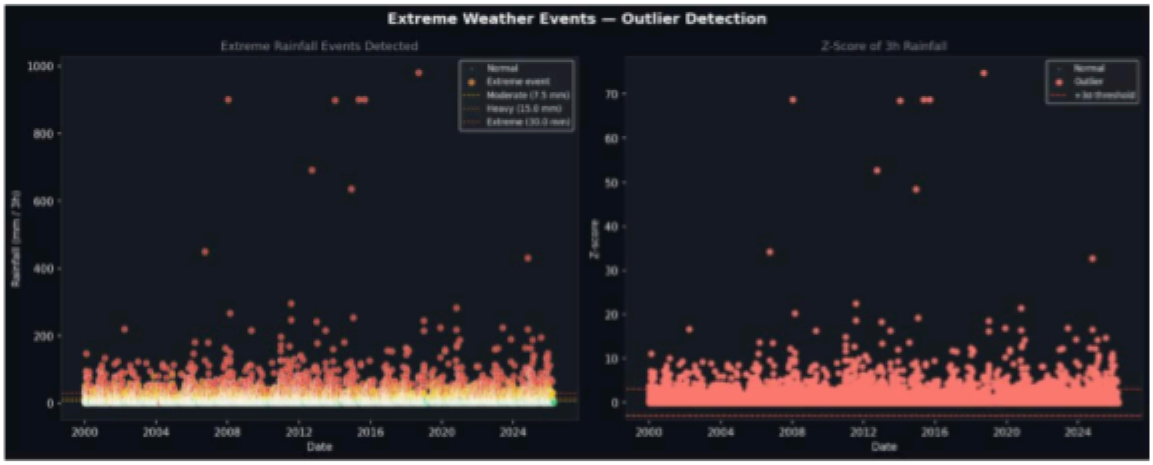

Model Performance Comparison
Metric Comparison
Score
0.999 0.999 1.000
1.000 1.000 1.000
0.983 0.983 1.000
Accuracy
F1 (wtd)
AUC (OvR)
Random Forest
Gradient Boost
Logistic Regression
Random Forest — Confusion Matrix
Actual
Predicted
LOW
MOD
HIGH
EXT
10203
3175
1228
694


Albay Flood Risk Prediction — Summary Dashboard
99.9%
Accuracy
99.9%
F1 Score
0.9999
AUC (OvR)
Per-Class Metrics (Random Forest)
Precision
Recall
F1
1.0
0.8
0.6
0.4
0.2
0.0
LOW
MOD
HIGH
EXT
Records per Risk Class
Count
50000
40000
30000
20000
10000
0
51,057
15,877
6,140
3,470
LOW
MOD
HIGH
EXT


Random Forest — Top Feature Importances
rain_24h
rain_48h
rain_12h
rain24h_98446
rain_lag_12h
rain_lag_3h
rain24h_98440
rain24h_98543
rain_lag_6h
rain12h_98446
humidity_x_rain
rain_3h
rain_zscore
rain24h_98536
rain24h_98427
0.4530
0.1785
0.1043
0.0311
0.0238
0.0200
0.0199
0.0183
0.0175
0.0118
0.0110
0.0097
0.0094
0.0078
0.0073
0.0
0.1
0.2
0.3
0.4
Importance


Albay (Station 98444) — Rainfall Timeline & Flood Risk
3-Hour Rainfall
mm / 3h
Rain 3h (mm)
Extreme event
24-Hour Accumulated Rainfall
mm / 24h
Rain 24h (mm)
Moderate (15 mm)
High (50 mm)
Extreme (100 mm)
Predicted Flood Risk Class
Flood Risk
EXT
HIGH
MOD
LOW


Flood Risk Distribution — Albay
Overall Risk Class Distribution
EXTREME
4.5%
HIGH
8.0%
MODERATE
20.7%
LOW
66.7%
Mean Flood Risk by Year × Month
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
EXT
HIGH
MOD
LOW



datetime	station_id	rain_3h	rain_12h	rain_24h	rain_48h	temp	pressure	humidity	flood_risk	flood_risk_	extreme_e	extreme_ti	rain_zscore
1/1/2000	98444	51.0	51.0	51.0	51.0	24.8	1011.7	98.8	2	HIGH	True	catastrophic	3.70180000000000
1/1/2000 3:00	98444	3.0	54.0	54.0	54.0	26.2	1011.1	92.6	2	HIGH	True	extreme	###################
1/1/2000 6:00	98444	9.0	63.0	63.0	63.0	28.0	1008.5	86.8	2	HIGH	True	extreme	###################
1/1/2000 9:00	98444	0.0	63.0	63.0	63.0	28.2	1008.7	84.3	2	HIGH	False	normal	###################
1/1/2000 12:00	98444	0.0	12.0	63.0	63.0	27.1	1011.2	86.2	2	HIGH	False	normal	###################
1/1/2000 15:00	98444	0.0	9.0	63.0	63.0	25.4	1011.5	93.6	2	HIGH	False	normal	###################
1/1/2000 18:00	98444	0.0	0.0	63.0	63.0	26.2	1009.4	82.1	2	HIGH	False	normal	###################
1/2/2000	98444	10.0	10.0	73.0	73.0	25.7	1011.4		2	HIGH	True	extreme	0.56155300000000
1/2/2000 3:00	98444	0.1	10.1	22.1	73.1	29.2	1011.7	75.4	1	MODERATE	False	normal	###################
1/2/2000 6:00	98444	28.0	38.1	47.1	101.1	25.5	1009.6	95.3	1	MODERATE	True	extreme	1.94019800000000
1/2/2000 9:00	98444	0.1	38.2	38.2	101.2	26.8	1009.0	88.3	1	MODERATE	False	normal	###################
1/2/2000 12:00	98444	0.0	################	38.2	101.2	25.6	1011.0	90.9	1	MODERATE	False	normal	###################
1/2/2000 15:00	98444	0.0	28.1	38.2	101.2	25.0	1011.7	90.8	1	MODERATE	False	normal	###################
1/2/2000 18:00	98444	0.0	0.1	38.2	101.2	25.2	1010.0	90.9	1	MODERATE	False	normal	###################
1/2/2000 21:00	98444	10.0	10.0	48.2	111.2	24.0	1009.7	97.0	1	MODERATE	True	extreme	0.56155300000000
1/3/2000	98444	38.0	48.0	76.2	149.2	24.8	1011.8	93.1	2	HIGH	True	catastrophic	2.70611200000000
1/3/2000 3:00	98444	4.0	52.0	80.1000000000000	102.2	25.8	1012.0	91.4	2	HIGH	True	extreme	0.10200500000000
1/3/2000 6:00	98444	4.0	56.0	56.1	103.2	27.0	1009.7	85.2	2	HIGH	True	extreme	0.10200500000000
1/3/2000 9:00	98444	0.4	46.4	56.4	###############	26.4	1009.6	89.3	2	HIGH	False	normal	###################
1/3/2000 12:00	98444	0.3	8.7	56.7	94.9	25.0	1011.4	89.2	2	HIGH	False	normal	###################
1/4/2000	98444	4.0	8.7	60.7	98.9	26.0	1012.4	83.6	2	HIGH	True	extreme	0.10200500000000
1/4/2000 3:00	98444	0.0	4.7	60.7	98.9	28.9	1012.7	74.0	2	HIGH	False	normal	###################
1/4/2000 6:00	98444	0.1	################	50.8	99.0	29.3	1010.1	73.6	2	HIGH	False	normal	###################
1/4/2000 9:00	98444	1.0	5.1	13.8	90.0	26.4	1009.7	91.5	0	LOW	True	extreme	###################

model	accuracy	f1_weightec	precision_w	recall_weigh	auc_ovr
Random Fo	0.9994	0.9994	0.9994	0.9994	0.9999
Gradient Bo	0.9997	0.9997	0.9997	0.9997	1.0
Logistic Reg	0.9832	0.9834	0.9838	0.9832	0.9997

datetime	station_id	rain_3h	rain_zscore	flood_risk_l	extreme_tier	temp	pressure	humidity
1/1/2000	98444	51.0	3.7018000000000000	HIGH	catastrophic	24.8	1011.7	98.8
1/1/2000 3:00	98444	3.0	0.025413000000000000	HIGH	extreme	26.2	1011.1	92.6
1/1/2000 6:00	98444	9.0	0.48496200000000000	HIGH	extreme	28.0	1008.5	86.8
1/2/2000	98444	10.0	0.5615530000000000	HIGH	extreme	25.7	1011.4	
1/2/2000 6:00	98444	28.0	1.9401980000000000	MODERATE	extreme	25.5	1009.6	95.3
1/2/2000 21:00	98444	10.0	0.5615530000000000	MODERATE	extreme	24.0	1009.7	97.0
1/3/2000	98444	38.0	2.706112000000000	HIGH	catastrophic	24.8	1011.8	93.1
1/3/2000 3:00	98444	4.0	0.1020050000000000	HIGH	extreme	25.8	1012.0	91.4
1/3/2000 6:00	98444	4.0	0.1020050000000000	HIGH	extreme	27.0	1009.7	85.2
1/4/2000	98444	4.0	0.1020050000000000	HIGH	extreme	26.0	1012.4	83.6
1/4/2000 9:00	98444	1.0	-00.1277690000000000	LOW	extreme	26.4	1009.7	91.5
1/4/2000 12:00	98444	1.0	-00.1277690000000000	LOW	extreme	25.1	1012.6	89.8
1/6/2000	98444	2.0	-00.0511780000000000	LOW	extreme	25.2	1011.6	95.3
1/6/2000 6:00	98444	1.0	-00.1277690000000000	LOW	extreme	28.6	1009.0	80.0
1/7/2000	98444	1.0	-00.1277690000000000	LOW	extreme	23.6	1012.3	97.0
1/10/2000	98444	20.0	1.3274670000000000	MODERATE	extreme	24.2	1013.6	94.2
1/10/2000 18:00	98444	1.0	-00.1277690000000000	MODERATE	extreme	26.4	1012.9	83.6
1/11/2000	98444	1.0	-00.1277690000000000	MODERATE	extreme	26.4	1012.9	83.6
1/12/2000 15:00	98444	2.0	-00.0511780000000000	LOW	extreme	25.4	1009.7	93.6
1/12/2000 18:00	98444	4.0	0.1020050000000000	LOW	extreme	25.2	1008.3	95.3
1/12/2000 21:00	98444	4.0	0.1020050000000000	LOW	extreme	24.5	1008.0	98.8
1/13/2000	98444	8.0	0.4083700000000000	MODERATE	extreme	24.8	1009.6	98.8
1/13/2000 3:00	98444	15.0	0.9445100000000000	MODERATE	extreme	25.7	1010.3	93.1

Challenges and Solutions

Challenge 1: Class Imbalance in the Risk Labels

Within the initial runs, the model kept predicting LOW risk for almost every observation while still producing high accuracy scores. The group realized that this happened because most records in the dataset naturally belonged to the LOW-risk category.

Solution: In order to give minority classes like HIGH and EXTREME more weight during training, the group added `class_weight='balanced'` to both the Random Forest and Logistic Regression models. To guarantee that every risk class was fairly represented in both the training and testing datasets, `stratify=y` was also added to the `train_test_split` function.

Challenge 2: Aligning Multiple Station Timestamps

The supporting stations did not always have matching timestamps with the main Albay station. Directly merging the datasets resulted in many NaN values and misaligned rows, which affected the rolling window calculations.

Solution: In the `engineer_features` function, each supporting station dataset was resampled into a consistent 3-hour interval using `.resample('3h').mean()`. The datasets were then re-indexed to match the datetime index of the Albay station. This ensured proper alignment before applying rolling and lag operations.

Challenge 3: Feature Matrix Becoming Too Large

After adding rainfall, pressure, and humidity features from all six supporting stations, the feature matrix became significantly larger. The group became concerned that this could slow down training or lead to overfitting.

Solution: The Random Forest parameters max_depth=12 and min_samples_leaf=5 helped limit model complexity despite the expanded feature set. Feature importance analysis was also used to confirm that the model was not relying on irrelevant variables. In addition, the 5-fold cross-validation results showed that the model generalized well instead of memorizing the training data.

1. Conclusion

All planned tasks for this week were successfully completed. The group selected Classification as the most suitable approach, developed a complete multi-station flood risk prediction pipeline, and fine-tuned the models for better performance. Running the pipeline on the full dataset confirmed that the approach was highly effective, with all three classifiers achieving accuracy above 98%, while Gradient Boosting and Random Forest both exceeded 99.9%.

One major finding from the results was that accumulated rainfall over longer periods, particularly 24-hour and 48-hour windows, was much more predictive than short-term or instantaneous rainfall readings. These features accounted for more than 63% of the model's decision-making based on feature importance analysis. The inclusion of neighboring station data, especially from Catanduanes (98446), also contributed significantly to the model's performance and validated the effectiveness of the multi-station design. Lastly, the outlier The detection module added practical value by identifying historically extreme rainfall events that a standard classifier alone would not easily highlight.